

Self-Organized Criticality: Sandpile Model on Undirected Graph, Influence of Topology and Dissipation Rules on Avalanches

1. Introduction

Almost everything in nature tends to cluster around a center value. An example might be the height of humans. This is described by a normal distribution which itself got its name because this pattern is so often observed not only in nature but in human-made systems as well.

Contrary to this, there are systems which could not be described by normal distribution. One example are the forest fire sizes. If they were to adhere to normal distribution, there would be known an *average forest fire size*. Even though most of forest fires are small, there is nothing limiting the potential size. This is one of key properties of systems better described by power-law, scale-invariance.

Normal distribution is *heavy tailed*, meaning that most of values are distributed within small range around the center (mean) and values far from it are very improvable. The probability density function of normal distribution is described in Equation 1

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/(2\sigma^2)} \quad (1)$$

μ ... mean
 σ^2 ... variance

TODO: plot normal dist?

As opposed to this, power-law¹ is *long tailed*. Its exponential character described in Equation 2 result in majority of the values being very far. That corresponds with the scale-invariance: Any (forest fire) size is realistically possible, even though smaller sizes are more likely. Power-law exhibit linear relationship between $\log f(x)$ and $\log x$. In a $\log$ - $\log$ plot it forms a straight line.

$$f(x) = ax^{-k} \quad (2)$$

k ... constant exponent

TODO: plot power-law? in normal and log-log plot

The emergence of power-law distribution is often connected to systems critical state. Consider phase transition of water from liquid to vapor. At the liquid-vapour boundary curve, both liquid and vapour can coexist but are distinctly separate. The boundary terminates at some critical temperature and critical pressure at which the critical point of a system is defined. While the system is in the critical state, the water is in not in any one state.

Systems in critical state are most responsive to input and behaves unpredictable. This is the case for phase transition and forest fires. Although, there is a significant difference. Criticality can be reached in many systems by fine-tuning certain parameters. In the case of phase transition the parameters were pressure and temperature. However, systems exist which reach criticality independently of parameters.

TODO NOTE?: usually high degree of freedom systems

¹Power-law on $(0, \infty)$ cannot be a probability distribution, because the areas near 0 and under the tail are infinite. However, with simple restrictions: $f(x) = ax^{-k}$, $k > 1$, $x > x_{\min}$ it is a valid distribution.

The phenomenon of systems reaching criticality by themselves is called *self-organized criticality* (SOC). The concept of SOC was first discovered in 1987 by Bak, Tang and Wiesenfeld and its properties were demonstrated on a sandpile model [1]. This model and its properties is the main focus of this thesis.

TODO: talk about the idea of putting it on random graphs, not just general?

The original sandpile model was a cellular automaton on a square and cubic lattice. However, the idea is to use another underlying graph structures for the automaton. Before this generalization we introduce few needed concepts from graph theory.

2. Graph Theory

An undirected graph is defined as a pair, containing set of vertices and set of edges. The graph has n vertices and m edges.

$$\begin{aligned} G &= (V, E) \\ V &= \{v_1, v_2, \dots, v_n\} \\ E &\subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}, \quad |E| = m \end{aligned} \quad (3)$$

The neighbourhood of a vertex is a set of vertices connected to it via an edge.

$$\text{nei}(v_i) = \{v_j \mid \{v_i, v_j\} \in E\}, \quad (4)$$

The vertex degree is defined as the size of vertex neighbourhood.

$$d(v_i) = |\text{nei}(v_i)| \quad (5)$$

Path between two vertices v_i and v_j is a sequence of vertices beginning in the first and ending in the second where every two vertices in path must be directly connected.

$$\begin{aligned} P &= (v_1, v_2, \dots, v_k), \quad v_1 = v_i, \quad v_k = v_j, \\ \{v_l, v_{l+1}\} &\in E \quad \forall l \in \{1, \dots, k-1\} \end{aligned} \quad (6)$$

If there exist no such path, let $P = ()$.

Similarly, path length is the number of *steps* (edges) needed to walk the path. If path doesn't exist we define its length as zero.

$$\text{len}(P) = \begin{cases} |P| - 1 & |P| \geq 2 \\ 0 & \text{else} \end{cases} \quad (7)$$

The shortest path between vertices v_i and v_j is the path with minimal length among all paths connecting these vertices. Let $\mathbb{P}(v_i, v_j)$ be the set of all paths between v_i and v_j :

$$P_{ij}^* = \min_{P \in \mathbb{P}(v_i, v_j)} \text{len}(P) \quad (8)$$

A graph has an average (shortest) path length $\text{len}(G)$. Let $\mathbb{P}^*$ be the set of all shortest paths:

$$\text{len}(G) = \frac{1}{n(n-1)} \sum_{i \neq j} \text{len}(P_{ij}^*) \quad (9)$$

TODO: small-world network properties (the previous needed for that)

3. Sandpile Model

The very original sandpile model is a cellular automaton on a lattice. Every cell can hold up to K grains, the height of a cell is a function of position $z(x, y, \dots)$. Fixed boundary conditions are used, on a boundary $z = 0$ cannot be changed. In two dimensions z is updated as follows:

$$\begin{aligned} z(x, y) &\rightarrow z(x, y) - 4 \\ z(x \pm 1, y) &\rightarrow z(x \pm 1, y) + 1 \\ z(x, y \pm 1) &\rightarrow z(x, y \pm 1) + 1 \end{aligned} \tag{10}$$

if z exceeds (or meets) critical value $K = 4$. The system is initialized randomly but with $z \gg K$. That initial state or rather the size of a lattice along with $\mathbb{Z}$ (the height of all cells) is called *configuration*.

For simplicity we can imagine a simple chessboard and falling sand grains. Each grain falls on a chess square determined by a random distribution². If the number of grains on any square reaches four, the sandpile topples and the four grains fall on adjacent squares, possibly causing additional squares to topple. In case the square does not have four adjacent squares (i.e. it is on the edge of chessboard) the remaining grains fall off the board and are therefore removed from the model.

We say that cell (square) is stable if it contains less than K grains. The system is stable if all of its cells are stable. If a grain is added to any stable cell, it can destabilize it resulting in an avalanche. Avalanche is the set of topplings which occur before the system is stabilized again.

The sandpile model is a dissipative dynamical system. The dissipation is demonstrating on the boundaries. When a grain leaves the chessboard (or enter a boundary cell) it dissipated from the system. Dissipation is an important property of the model. Without dissipation the model supersaturates, leading to one infinite avalanche.

We can express the square lattice (chessboard) as a graph $G = (V, E)$ with size³ $\sqrt{n}$. For a boundary condition, the boundary vertices height is fixed to zero $z = 0$. Alternatively, we may imagine only one vertex, called *sink vertex*, with such condition. This sink is connected to all boundary vertices, thus creating the same effect. Please note, that this simplification creates a multigraph, because corner vertices and sink are connected via two edges. For the rest of this thesis we will consider the simplified version with one sink vertex and therefore a multigraph. However, it is trivial to create multiple sink vertices to restore a simple undirected graph.



Figure 1: Comparison of square lattice sink vertices (blue). On the left is a graph with sinks at the boundary - as in the original model. On the right is a multigraph, square lattice with only one sink connected to all vertices at the boundary which is hinted by outgoing edges.

All properties of the cellular automaton system holds on a square lattice. This model was later generalized from square lattice to arbitrary graph [2]. That generalization is important because it allows us to study the models behaviour for different graph topologies.

²In the original model, all sand grains were distributed initially. It might be more obvious if we let grains fall one at a time for demonstration purposes.

³In a square lattice, the size² = n , which is the number of vertices in graph.

3.1. Graph Topologies

Standard choice for graph in sandpile model is square lattice with arbitrary size.

TODO: square lattice - properties

TODO: ER graph - we don't have

TODO: WS graph - we have

3.2. Dissipation Rules

With the introduction of arbitrary graph we must reconsider dissipation rules. The square lattice has explicit boundaries which serves as sinks. However, many graphs do not even have an implicit boundary. The existence of sink is fundamental for the model not to supersaturate.

A dissipation rule determines how many times a vertex should be connected to the sink. There is not a single rule which would be generally applied to any graph. We will discuss different rules, their properties and analogies.

As discussed in Section 3 the existence of a sink vertex is necessary for any graph representation of a sandpile model. However there are various rules by which you can determine if a given vertex should be connected to a sink vertex and how many times if so.

For example the original model uses a rule (named by me as) *Fill to Four*. This rule states that the degree of vertex must be exactly four. The sink vertex is connected to all vertices as many times it is needed to fulfill that rule. For the square lattice, only vertices on the edge are connected (exactly once except for corner vertices, which are connected twice). This rule precisely reproduces the original chessboard idea.

For an arbitrary graph though, the rule is insufficient. It does not ensure dissipation which can lead to graph supersaturation and never-ending avalanche. This rule does not reflect on the vertex degree if it is higher than four.

One possible rule which guarantees dissipation on any graph may be *All Once*. The rule is fair in a sense that all vertices have connection to the sink and is uniform because every vertex, no matter its degree, is connected exactly once.

Finally, another rule which assures dissipation is *As Many As Neighbours*. This rule connects every vertex to sink proportionally to the number of its neighbouring vertices, therefore emphasizing how many connections it has.

Every rule represents a different approach and not all can be applied to all types of graph topology.

4. Observed Behaviour

TODO: how did it work for different types of graphs and dissipation rules? also the graph configurations (eg β for WS)

TODO: metrics, how it changed criticality? origin distribution, avalanche size (max, avg)

5. Technical Documentation

TODO

Bibliography

- [1] P. Bak, C. Tang, and K. Wiesenfeld, "Self-organized criticality: An explanation of the $1/f$ noise," *Physical Review Letters*, vol. 59, no. 4, pp. 381–384, July 1987, doi: 10.1103/PhysRevLett.59.381.

- [2] A. E. Holroyd, L. Levine, K. Mészáros, Y. Peres, J. Propp, and D. B. Wilson, “Chip-Firing and Rotor-Routing on Directed Graphs,” in *In and Out of Equilibrium 2*, Birkhäuser Basel, 2008, pp. 331–364. doi: 10.1007/978-3-7643-8786-0_17.

TODO

- multigraph?

TODO PS

- oponent
- předběžná verze L. Vášovi